

Windows + CUDA Installation Guide for difflogic

Mahmoud Yassin

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Overview

This document provides a **fully tested and reproducible installation guide** for the `difflogic` library on **Windows** with **CUDA** acceleration.

Info

Direct installation via `pip install difflogic` currently **fails on Windows** due to NVC-C/MSVC incompatibilities and unsupported CUDA syntax. This guide documents the *only known working path*.

Tested Configuration

- **OS:** Windows 10 / 11 (x64)
- **Python:** 3.10
- **PyTorch:** 2.5.1 + CUDA 12.1
- **CUDA Toolkit:** 12.1
- **MSVC Toolchain:** Visual Studio Build Tools 2022 (v14.38)
- **GPU:** NVIDIA RTX series (tested on RTX 3050 Laptop GPU)

Known Limitations

Warning

The following configurations are known to fail.

- `pip install difflogic` on Windows
- MSVC versions newer than 14.38
- CUDA versions not matching PyTorch (e.g. cu118)
- Building from `main` without patches

Required Source Fix (Mandatory)

Successful Windows builds require the following commit:

```
81347af (Update difflogic_kernel.cu)
```

Info

This commit fixes **CUDA designated-initializer syntax** that fails under NVCC on Windows.

Upstream commit link:

<https://github.com/Felix-Petersen/difflogic/commit/81347af>

Installation Procedure

1. Install Visual Studio Build Tools 2022

Download from:

https://aka.ms/vs/17/release/vs_buildtools.exe

Select **only**:

- MSVC v143 (C++ build tools)
- MSVC version **14.38.x**
- Windows 10/11 SDK

Warning

Do **not** install newer MSVC versions. NVCC will reject them.

2. Create Conda Environment

Open **Anaconda Prompt**:

```
conda create -n difflogic2 python=3.10 -y
conda activate difflogic2
```

3. Install PyTorch (CUDA 12.1)

```
pip install torch torchvision torchaudio \
--index-url https://download.pytorch.org/whl/cu121
```

Verify:

```
python -c "import torch; print(torch.__version__, torch.version.cuda, torch.cuda.is_available())"
```

4. Install CUDA Toolkit 12.1

Download from:

<https://developer.nvidia.com/cuda-12-1-0-download-archive>

Verify:

```
nvcc --version
```

Expected:

```
Cuda compilation tools, release 12.1
```

5. Load Correct MSVC Toolchain

Inside **Anaconda Prompt**:

```
"%ProgramFiles(x86)%\Microsoft Visual Studio\2022\BuildTools\Common7\Tools\VsDevCmd.bat" ^  
-arch=x64 -vcvars_ver=14.38
```

Set environment variables:

```
set DISTUTILS_USE_SDK=1  
set MSSdk=1
```

6. Clone Repository and Checkout Fix

```
git clone https://github.com/Felix-Petersen/difflogic.git  
cd difflogic  
git checkout 81347af
```

Required Source Patch (Explained)

Problem. NVCC on Windows fails on union designated initializers:

```
const union scalar_t_ val = {.signed_scalar = x};
```

Fix (NVCC-safe).

```
union scalar_t_ val;  
val.signed_scalar = x;
```

If const is required:

```
const union scalar_t_ val = [] __device__ (auto x) {  
    union scalar_t_ tmp;  
    tmp.signed_scalar = x;  
    return tmp;  
}(value);
```

Build and Install

```
python -m pip install -v --no-build-isolation --no-cache-dir .
```

Success

If successful, you will see:

```
Successfully built difflogic  
Successfully installed difflogic-0.1.0
```

Windows DLL Loading Fix (Python Method)

Info

On Windows, CUDA extensions require explicit DLL path registration.

```
import os
import torch

torch_pkg_dir = os.path.dirname(torch.__file__)
torch_lib_dir = os.path.join(torch_pkg_dir, "lib")

os.add_dll_directory(torch_lib_dir)
os.add_dll_directory(r"C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v12.1\bin")

import difflogic_cuda
print("difflogic_cuda loaded OK")
```

Final Verification

```
python -c "import difflogic, difflogic_cuda, torch; print(torch.cuda.get_device_name(0))"
```

Status Summary

- Windows: [Supported](#)
- CUDA: [Enabled](#)
- PyTorch 2.5: [Supported](#)
- NVCC: [Compatible](#)
- Installation: [Reproducible](#)